MALAPPURAM SAHODAYA APTITUDE TEST
EXAMINATION 2023-24
DISTRICT LEVEL

Class: 12(PCB) **Date: 4/11/23** **Marks: 100**

Name of the Student: **Time: 2 Hrs**

General Instructions:

1. Candidate will be supplied with separate Question Paper and Answer Sheet.
2. Answer the questions in the OMR sheet by shading the appropriate bubble.
3. Shade the bubbles with black ball pen/ HB pencil.
4. Question number ***1 to 100*** *one mark each.*
5. For each question below, four options are given. One of them is correct answer. Make your choice and shade the correct box.

1. Which organelle is responsible for the synthesis of lipids and detoxification of drugs in a cell?
a. Mitochondria b. Endoplasmic reticulum c. Nucleus d. Golgi apparatus Answer:

2. What is the process by which plants lose water vapor to the atmosphere through small openings in their leaves called?
a. Transpiration b. Photosynthesis c. Respiration d. Digestion

3. Which enzyme is responsible for breaking down starch into maltose in the digestive system?
a. Amylase b. Pepsin c. Lipase d. Trypsin

4. In which phase of the cell cycle does DNA replication occur?
a. G1 phase b. S phase c. G2 phase d. M phase

5. Which of the following hormones is produced by the pancreas and regulates blood sugar levels?
a. Insulin b. Thyroxine c. Adrenaline d. Estrogen

6. What is the process by which white blood cells engulf and destroy foreign invaders such as bacteria and viruses?
a. Phagocytosis b. Osmosis c. Diffusion d. Active transport

7. Which of the following is a component of the ecosystem's abiotic factors?
a. Producers b. Consumers c. Temperature d. Decomposers ‘

8. What is the primary function of the nephron in the kidney?
a. Filtration of blood b. Production of urine
c. Digestion of proteins d. Synthesis of hormones

9. Which gas is released by plants during respiration and is also a waste product of animal respiration?
a. Oxygen b. Carbon dioxide c. Nitrogen d. Hydrogen

10. Which of the following is a genetic disorder caused by the presence of an extra chromosome 21?
a. Hemophilia b. Down syndrome c. Cystic fibrosis d. Huntington's disease

11. What is the name of the hormone that stimulates milk production in the mammary glands of females after childbirth?
a. Estrogen b. Progesterone c. Prolactin d. Testosterone

12. Which part of the human brain is responsible for regulating vital functions such as heart rate and breathing?
a. Cerebrum b. Cerebellum c. Medulla oblongata d. Hypothalamus

13. Which of the following is an example of a non-communicable disease?
a. Influenza b. Tuberculosis c. Diabetes d. Malaria

14. What is the name of the process by which plants convert light energy into chemical energy in the form of glucose?
a. Respiration b. Fermentation c. Photosynthesis d. Transpiration

15. Which of the following is responsible for the breakdown of red blood cells and the recycling of their components?

a. Liver b. Spleen c. Kidneys d. Pancreas

16. What is the process by which a cell engulfs large particles or even other cells by wrapping pseudopodia around them?

a. Pinocytosis b. Exocytosis c. Phagocytosis d. Endocytosis

17. Which of the following is the male sex hormone responsible for the development of secondary sexual characteristics?

a. Estrogen b. Progesterone c. Testosterone d. Prolactin

18. What is the term for the process by which an organism maintains a stable internal environment despite external fluctuations?

a. Adaptation b. Homeostasis c. Metabolism d. Evolution

19. Where does the digestion take place in acidic medium in the digestive system?

a. Stomach b. Small intestine c. Large intestine d. Pancreas

20. What is the name of the process by which DNA serves as a template for the synthesis of RNA?

a. Transcription b. Translation c. Replication d. Mutation

21. Ferrous oxide has a cubic structure and each edge of the unit cell is 5.0 A^0. Assuming the density of the oxide as 4.0 g cm^{-3}, find the number of Fe^{2+} and O^{2-} ions present in each unit cell.

a. four Fe^{2+} and four O^{2-}
c. four Fe^{2+} and two O^{2-}
b. two Fe^{2+} and four O^{2-}
d. three Fe^{2+} and three O^{2-}

22. Each of the following solids show, the Frenkel defect except

a. ZnS b. AgBr c. AgI d. KCl

23. The mole fraction of the solute in one molal aqueous solution is

a. 0.009 b. 0.018 c. 0.027 d. 0.036

24. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is

a. $P_3 > P_1 > P_2$ b. $P_3 > P_2 > P_1$ c. $P_2 > P_1 > P_3$ d. $P_1 > P_2 > P_3$

25. Which of the following solutions will have lowest elevation in boiling point?

a. 0.1m KCl b. 0.05m NaCl c. 1m $AlPO_4$ d. 0.1m $MgSO_4$

26. The weight of silver displaced by a quantity of electricity which displaces 5600ml of O_2 at STP will be

a. 5.4g b. 10.8g c. 54.9g d. 108.0g

27. On electrolysis of dilute sulphuric acid using Platinum (Pt) electrode, the product obtained at the anode will be:

a. Oxygen b. H_2S gas c. SO_2 gas d. Hydrogen gas

28. A substance 'A' decomposes by a first-order reaction starting initially with [A] = 2.00M and after 200min, [A] becomes 0.15M. For this reaction t1/2 is

a. 53.72 min b. 50.49 min c. 48.45 min d. 46.45 min

29. For a certain reaction, the rate = k $[A]_2[B]$, when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

a. increases by a factor of six
c. increases by a factor of three
b. increases by a factor of nine
d. decreases by a factor of nine

30. The volume of gases H_2, CH_4, CO_2 and NH_3 adsorbed by 1 gm charcoal at 293 K can be given in the order?

a. $CH_4 > CO_2 > NH_3 > H_2$
c. $NH_3 > CO_2 > H_2 > CH_4$
b. $CO_2 > NH_3 > H_2 > CH_4$
d. $NH_3 > CO_2 > CH_4 > H_2$

31. Pumice stone is an example of-

a. Foam b. Sol c. Gel d. Solid sol

32. Cassiterite is the ore of which metal?

a. Hg b. Sb c. Sn d. Ni

33. Write the IUPAC name of the compound

a. 6–bromo–7–chloro–oct–5–en–2–yne
b. 3–bromo–2–chloro–oct–3–en–6–yne
c. 2–chloro-3–bromo–oct–3–en–6–yne
d. 2–chloro-3–bromo–oct–6–yn–3–ene

34. Water samples with BOD values of 4ppm and 18ppm, respectively, are:

a. highly polluted and clean
b. highly polluted and highly polluted
c. clean and highly polluted
d. clean and clean

35. CH_3COONa is a/an

a. Acidic salt
b. Basic salt
c. Neutral
d. Amphoteric

36. An inductor coil with an internal resistance of 50 ohm stores magnetic field energy of 180 mJ and dissipates energy as heat at the rate of 200 W when a constant current is passed through it. The inductance of the coil will be:

a. 90 mH
b. 120 mH
c. 45 mH
d. 30 mH

37. A metallic cylindrical wire 'A' has length 10 cm and radius 3 mm. Another hollow cylindrical wire 'B' of the same metal has length 10 cm, inner radius 3 mm and outer radius 4 mm. The ratio of the resistance of the wires A to B is:

a. $\frac{7}{9}$
b. $\frac{9}{7}$
c. $\frac{9}{16}$
d. $\frac{9}{16}$

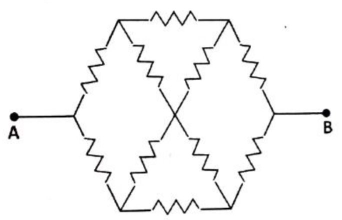
38. In the magnetic meridian of a certain plane, the horizontal component of earth's magnetic field is 0.36 Gauss and the dip angle is 60°. The magnetic field of the earth at the location is:

a. 0.72 Gauss
b. 0.18 Gauss
c. 0.42 Gauss
d. 0.56 Gauss

39. A current carrying long solenoid is formed by winding 200 turns per cm. If the number of turns per cm is increased to 201 keeping the current constant, then the magnetic field inside the solenoid will change by:

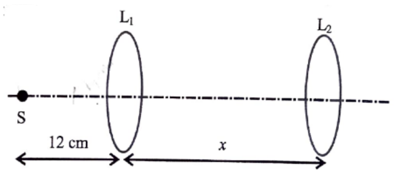
a. 0.2 %
b. 0.4 %
c. 0.5 %
d. 1 %

40. Find the effective resistance between points A and B. Each resistance is equal to R.



a. 2R
b. $\frac{3}{4}$R
c. 3R
d. $\frac{4}{3}$R

41. Two thin convex lenses L_1 and L_2 have focal lengths 4 cm and 10 cm, respectively. They are separated by a distance of x cm as shown in the figure. A point source S is placed on the principal axis at a distance 12 cm to the left of L. If the image of S is formed at infinity, the value of x is



a. 6
b. 16
c. 14
d. 24

42. A thin convex lens of refractive index 1.5 has a focal length of 10 cm in air. When the lens is immersed in a fluid, its focal length becomes 70 cm. The refractive index of the fluid is:

a. 1.33
b. 1.6
c. 1.25
d. 1.4

43. In a Young's double slit experiment which of the following statements is NOT true?

a. Angular separation of the fringes remains constant when the screen is moved away from the plane of the slits.
b. Fringe separation increases when the separation between the two slits decreases.
c. Sharpness of the fringe pattern decreases when the source slit width increases.
d. Distance between the fringes decreases when the separation between slits and the Screen increases.

44. N capacitors, each with 1 F capacitance, are connected in parallel to store a charge of 1C. The potential across each capacitor is 100 V. If these N in series, the capacitors are now connected equivalent capacitance in the circuit will be:
   a. 10^{-4} F b. 10^{-6} F c. 10^{-10} F d. 5×10^{-8} F
45. An unpolarised light is incident on a glass slab such that the reflected ray is totally polarised. If the angle of refraction is 30°, the refractive index of the glass is:
   a. 1.5 b. 1.73 c. 1.41 d. 1.45
46. A hollow sphere of radius 'r' encloses an electric dipole composed of two charges +q and -q. The net flux of electric field through the surface of the sphere due to the enclosed dipole is:
   a. $\frac{2q}{\varepsilon_0}$ b. $\frac{2q}{\varepsilon_0}.4\pi r^2$ c. zero d. $\frac{q}{\varepsilon_0}$
47. A series LCR circuit consists of a variable capacitor connected to an inductor of inductance 50 mH, resistor of resistance 100 ohm and an AC source of angular frequency 500 rad/s. The value of capacitance so that maximum current may be drawn into the circuit is:
   a. 60 µF b. 50 µF c. 100 µF d. 80 µF
48. In a current carrying coil of inductance 60 mH, the current is changed from 2.5 A in one direction to 2.5 A in the opposite direction in 0.10 sec. The average induced EMF in the coil will be:
   a. 1.2 V b. 2.4 V c. 3.0 V d. 1.8 V
49. A long wire carrying ac current of 5 A lies along the positive z-axis. The magnetic field at the point with position vector $\vec{r} = \left(\hat{\imath} + 2\hat{\jmath} + 2\hat{k}\right)$ m will be: ($\mu_0 = 4\pi \times 10^{-7}$ in SI units)
   a. $2\sqrt{5} \times 10^{-7}\,T$ b. $5 \times 10^{-7}\,T$ c. $0.33 \times 10^{-7}\,T$ d. $0.66 \times 10^{-7}\,T$
50. What is the de Broglie wavelength corresponding to a ball of mass 100 g moving with a speed of 33 m/s? (Planck's constant = 6.6×10^{-34} J/s)
   a. 1×10^{-34} m b. 2×10^{-34} m c. 3×10^{-34} m d. 4×10^{-34} m
51. Which country's new rules on drones, permit them to fly over people and at night in the country?
   [A] China [B] Russia [C] United States [D] South Korea
52. As a part of Mission SAGAR-III, Indian Naval Ship INS Kiltan as reached which country to deliver relief materials?
   [A] Madagascar [B] Philippines [C] Cambodia [D] Laos
53. India's first pollinator park has been established in which state?
   [A] Haryana [B] Uttarakhand [C] Himachal Pradesh [D] Sikkim
54. Which state has been declared a "Disturbed area" for six more months, by the Home Ministry?
   [A] Assam [B] Nagaland [C] Mizoram [D] Manipur
55. Two Industrial Corridor nodes at Krishnapatnam and Tumakuru, have been approved under which Industrial corridor?
   [A] Amritsar-Kolkata Industrial Corridor [C] Chennai-Bengaluru Industrial Corridor
   [B] Bengaluru-Mumbai Industrial Corridor [D] Vizag-Chennai Industrial Corridor
56. National Pension System (NPS) is managed by which organisation?
   [A] IRDAI [B] PFRDA [C] Finance Ministry [D] RBI
57. India asked which country to assist its crew members of the stranded cargo vessels MV Jag Anand and MV Anastasia?
   [A] Sri Lanka [B] China [C] Japan [D] Thailand
58. Lal Pahari, where the first hilltop Buddhist monastery of the Gangetic Valley has been found, is in which state?
   [A] Bihar [B] Uttar Pradesh [C] Jharkhand [D] West Bengal
59. What is the name of the webinar serried launched by the Commerce Ministry, along with NPC and QCI?
   [A] Atmanirbhar Industries [C] MSME Marathon
   [B] Udyog Manthan [D] Self-Reliance Series
60. Which organisation recently deployed its all-women team for contingency duties?

[A] Central Armed Police Force   [C] National Disaster Response Force
[B] Indo-Tibetan Border Police   [D] Central Industrial Security Force

61. Rs.1200 divided among P, Q and R. P gets half of the total amount received by Q and R. Q gets one-third of the total amount received by P and R. Find the amount received by R ?
a. Rs. 500   b. Rs. 1200   c. Rs 700   d. Rs. 1100

62. A, B and C together start a business. B invests of 1/6 of the total capital while investments of A and C are equal. If the annual profit on this investment is 33600, find the difference between the profits of B and C ?

a. 6000   b. 7200   c. 9600   d. 8400

63. Rs. 700 is divided among A, B, C so that A receives half as much as B and B half as much as C. Then C's share is ?

a. Rs. 400   b. Rs. 500   c. Rs 200   d. Rs. 300

64. Who among the following defines intelligence as the aggregate or global capacity of the individual to act purposefully, think rationally and deal effectively with the environment?
A) Spearman   C) David Wechsler
B) Binet   D) Gallon

65. Who among the following was the first person that devised systematic tests to measure intelligence of children?
A) Terman   C) Thorndike
B) Binet   D) Wechsler

66. Which of the following expressions is more formal?
(A) Merry X-mas from the team at Silverzone's.
(B) Like our furniture? Make it yours with colours and fabrics that will blow you away!
(C) We're supper happy that you could be with us on our special day! Thanks again for the awesome gift.
(D) Thank you for confirming your appointment! We'll see you on Monday, 17 January at 10 a.m.

67.In the question given below are three statements followed by three conclusions numbered I, II and III. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follow/s from the given statements, disregarding commonly known facts.
Statements
I: No radio is a magazine.
II: No television is a magazine.
III: Some televisions are newspapers.
Conclusions
I: No radio is a television.
II: Some newspapers are not magazines.
III: All newspapers are definitely not televisions.
(A) All follow   (B) Only I and II follow
(C) Only II follows   (D) Only II and III follow

68. Find out which of the sentences given as options can fill up the gap between the given sentences in order to make the theme complete.
Mr. Raman is undoubtedly a competent professional.

-- So did the other professionals in the company.
i. Still he contributed significantly to the growth of the company.
ii. At times he has had serious differences with the chairman regarding the corporate policies.
iii. He was not considered for the post of the chairman.
(A) i only   (B) ii and iii only   (C) ii only   (D) i and iii only

69. Read the claim and the supporting evidence.
Claim: Parents should limit the sugar-sweetened beverages their children drink.
Evidence: Sugar-sweetened beverages can contribute to health problems such as heart disease and diabetes.
Why does the evidence support the claim? Choose the analysis that better explains the connection.
(A) Sugar sweetened beverages are often marketed to children through print and television advertisement.
(B) Parents are obliged to take steps to protect the health of their children.
(C) Sugar sweetened beverages are often addictive for the children.
(D) All of these

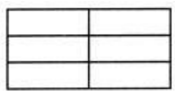
70. Change the voice.
Since he had not done the preliminary work, we had to cancel the meeting.
(A) Since the preliminary work had not been done by him, the meeting had to be cancelled.
(B) Since he had not done the preliminary work, the meeting had to be cancelled.
(C) The meeting was cancelled by us since he had not done the preliminary work.
(D) Both A and B

71. Arrange P, Q, R, S to make a correct sentence. An estimated 300, 000 Dalits had gathered
P: over the Peshwas, a faction of the Maratha empire
Q: the 1818 victory of the British East India Company
R: that ruled much of the subcontinent
S: in the Maharashtra village of Bhima Koregaon to celebrate before the British.
(A) PQRS (B) PRQS (C) SRQP (D) SQPR

72. Which of the following is a compound conjunction?
(A) Either-or (B) Even if (C) Both-and (D) Not only-but also

73. Over-extraction of groundwater can lead to various negative consequences. Which of the following is NOT a potential consequence of excessive groundwater extraction?
a. Increased soil fertility
b. Land subsidence
c. Saltwater Intrusion
d. Decreased baseflows in rivers

74. What is the number of rectangles in the figure given below?
a. 12
b. 11
c. 13
d. 14



75. How many vowels in the above arrangement are preceded by a symbol?
a. One b. Two c. Three d. Four

76. Which is the second element to the left of the sixth element from the right?
a. & b. @ c. N d. E

Directions(77-78): Find the missing element in the series given below:

77. ABD, EFH, IJL, MNP, QRT, ?
a. XYZ b. VWY c. WXZ d. UVX

78. ACE, ? ,MOQ,SUW
a. GIK b. FHJ c. GHJ d. GIL

79. Pointing to a photograph of a boy Mr.Ram said, “He is the son of the only son of my mother.” How is Mr Ram related to that boy?
a. Brother b. Uncle c. Cousin d. Father

80. Rita told Mani, “The girl I met yesterday at the beach was the youngest daughter of the brother-in-law of my friend’s mother.” How is the girl related to Rita’s friend?
a. Cousin b. Daughter c. Friend d. Aunt

81. What will be the code for ‘translate’?

a. #E8 b. @E9 c. @E8 d. #T9

82. What will be the code for ‘plan’?

a. %N5#E6 b. @E4 c. &N4 d. %N5

83. What does ‘%D4’ stands for?

a. sick b. turn c. point d. good

84. From the given options, find the pair which is similar to the given pair: 8:4

a. 27:9 b. 216:32 c. 72:24 d. 45:5

85. From the given options, choose the odd one out.

a. Bangladesh: Taka b. Brazil: Real c. Cyprus: Dollar d. Iran: Rial

Directions (Questions 86 & 87): Study the following information carefully and answer the questions given below:

Five persons - Amita, Babu, Chanda, Diwakar and Ishu - are travelling by a train. Amita is the mother of Chanda, who is the wife of Ishu. Diwakar is the brother of Amita and Babu is the husband of Amita.

86. How Babu is related to Ishu?

(A) Brother-in-law (B) Father-in-law (C) Father (D) Mother-in-law

87. How Amita is related to Ishu?

(A) Mother-in-law (B) Niece (C) Sister (D) Mother

88. A, B, C, D, E and F, not necessarily in that order, are sitting in six chairs regularly placed around a round table. It is observed that A is between D and F, C is opposite D, D and E are not on neighbouring chair. Which one of the following must be true?

(A) A is opposite B

(B) D is opposite E

(C) C and B are neighbours

(D) B and E are neighbours

89. Consider the series given below, and find the next term.

4 / 12 / 95, 1 / 1 / 96, 29 / 1 / 96, 26 / 2 / 96, ________.

(A) 24 / 3 / 96 (B) 25 / 3 / 96 (C) 26 / 3 / 96 (D) 27 / 3 / 96

90. Which letter will replace the question mark (?) in the following matrix?

D	M	V
F	P	Z
H	S	?

(A) D (B) C

(C) I (D) G

91. Study the following information and answer the given question.

Point B is 12 metre South of point A, Point C is 24 metre East of point B. Point D is 8 metre South of point C, point D is 12 metre East of point E and point F is 8 metre North of point E. Now, if a man is standing facing North at point C, how far and in which direction is point F?

(A) 12 m west (B) 24 m east (C) 12 m east (D) 24 m west

*Directions(Questions 92 & 93): In the following questions, the symbols @, #, %, $ and * are used with the following meanings as illustrated below.*

‘A @ B’ means ‘A is not smaller than B’

‘A # B’ means ‘A is neither smaller than nor equal to B’

‘A % B’ means ‘A is neither smaller than nor greater than B’

‘A $ B’ means ‘A is not greater than B’

‘A * B’ means ‘A is neither greater than nor equal to B’

Now in each of the following questions, assuming the given statements to be true, find the correct option for the given conclusions.

92. Statements: C#D, A@B, D*E, B%C

Conclusion: (1) A*E (2) A@E

(A) Only conclusion 1 is true

(B) Only conclusion 2 is true

(C) Either conclusion 1 or 2 is true

(D) Neither conclusion 1 nor 2 is true

93. Statements: I%J, G$H, H*I, H@K

Conclusion: (1) I*J (2) G@K

(A) Only conclusion 1 is true
(B) Only conclusion 2 is true
(C) Either conclusion 1 or 2 is true
(D) Neither conclusion 1 nor 2 is true

Directions (Questions 94 & 95): Study the following information to answer the given questions. A test with five sections namely Quant, Reasoning, English, Computer and GA has to be proofread in 6 hours where one hour needs to be spent on each section.

A break of one hour has to be taken in the third or the fourth hour. Proofreading cannot start with Quant and has to end with English. Computer has to immediately follow Reasoning with no break in between. Quant cannot be done immediately after Computers. Quant has to immediately follow GA with no break in-between.

94. In which hour the break is to be taken?

(A) Sixth (B) Fourth (C) Fifth (D) Third

95. Which section is to be proofread immediately after GA?

(A) English (B) Computers (C) Quant (D) Reasoning

Directions (Questions 96 & 97): Study the following information carefully to answer given questions.

A building has seven floors numbered one to seven, in such a way that the ground floor is numbered one, the floor above it is numbered two and so on such that the topmost floor is numbered seven. One out of seven people viz. Koyal, Naina, Dimple, Shruti, Payal, Arti and Anjali lives on each floor. Koyal lives on fourth floor. Payal lives on the floor immediately below Arti's floor. Arti does not live on the second or seventh floor. Dimple does not live on an odd numbered floor. Naina does not live on a floor immediately above or below Dimple's floor. Shruti does not live on the topmost floor. Anjali does not live on any floor below Payal's floor.

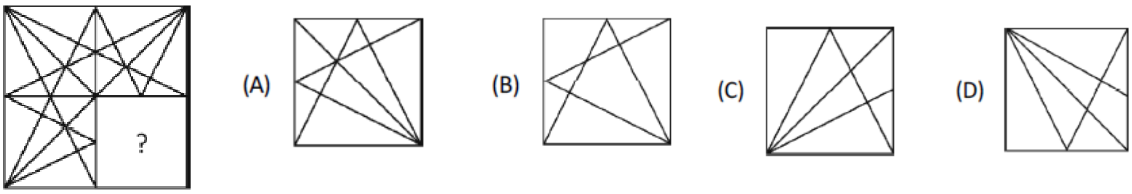
96. Who lives on the topmost floor?

(A) Dimple (B) Shruti (C) Anjali (D) Koyal

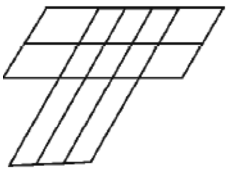
97. Who lives between Arti and Naina?

(A) Payal (B) Koyal (C) Shruti (D) Dimple

98. Which one of the alternative figures will complete the figure pattern?



99. How many parallelograms are there in this figure?



(A) 49
(B) 46
(C) 44
(D) 39

100. Consider the following statements.

1. Either A and B are of the same age or A is older than B.

2. Either C and D are of the same age or D is older than C.

3. B is older than C.

Which of the following conclusions can be drawn from the given statements?

(A) A is older than B
(B) B and D are of the same age
(C) D is older than C
(D) A is older than C

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